Def Let X be a set and $T: X \to X$ be a map. A **fixed pt** of T is a pt $x_* \in X$ s.t. $T(x_*) = x_*$.

Def Let (X,d), (Y,ρ) be metric spaces. A map $T: X \to Y$ is **Lipschitz w/ constant** $L \geq 0$, or **L-Lipschitz** for brevity, if

$$\forall x_1, x_2 \in X: \quad \rho(T(x_1), T(x_2)) \leq L\, d(x_1, x_2).$$

We say that T is **Lipschitz** if T is L-Lipschitz for some L. We say that T is a **contraction map** if T is L-Lipschitz for some $L \in [0,1)$.

Contraction mapping thm (aka, e.g., Banach fixed pt thm)

Let X be a complete metric space and $T: X \to X$ be a contraction map. Then T has a unique fixed pt $x_* \in X$. Moreover, $\forall x_0 \in X$, $\underbrace{T^n}_{\underbrace{T \circ T \circ \cdots T}_{n\text{-times}}}(x_0) \longrightarrow x_*$ as $n \to \infty$.

pf Let $L \in [0,1)$ be a Lipschitz const. for T.

Let $x_0 \in X$ be arbitrary. Define a sequence (x_n) in X by $x_n := T^n(x_0)$, so e.g. $x_n = T(x_{n-1})$.

Note that $\forall n \in \mathbb{N}$, we have

$$\begin{aligned} d(x_{n+1}, x_n) &= d(T(x_n), T(x_{n-1})) \\ &\leq L\, d(x_n, x_{n-1}) = L\, d(T(x_{n-1}), T(x_{n-2})) \\ &\leq L^2 d(x_{n-1}, x_{n-2}) \\ &\vdots \end{aligned}$$

So by finite induction,

$$d(x_{n+1}, x_n) \leq L^n d(x_1, x_0) \qquad \forall n \in \mathbb{N}.$$

Thus, for any $m, n \in \mathbb{N}$ w/ $m > n$,

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \cdots + d(x_{n+1}, x_n) \\ &\leq L^{m-1} d(x_1, x_0) + L^{m-2} d(x_1, x_0) + \cdots + L^n d(x_1, 0) \\ &= L^n d(x_1, x_0) \sum_{j=0}^{m-n-1} L^j \\ &\leq L^n d(x_1, x_0) \sum_{j=0}^{\infty} L^j \\ &= L^n d(x_1, x_0) \frac{1}{1-L}. \end{aligned}$$

WWTS (x_n) is Cauchy. Fix $\varepsilon > 0$. Since $0 \leq L < 1$,

$\exists N \in \mathbb{N}$ s.t. $\forall n \geq N$: $\quad L^n < \varepsilon \dfrac{(1-L)}{d(x_1, x_0)}$.

Then $\forall n, m \geq N$, the above yields

$$d(x_n, x_m) < \frac{\varepsilon \cancel{(1-L)}}{\cancel{d(x_1, x_0)}} \cancel{d(x_1, x_0)} \frac{1}{\cancel{1-L}} = \varepsilon .$$

$\Rightarrow$ (x_n) is a Cauchy sequence. Since (X, d) is complete, $\exists\, x_* \in X$ s.t. $x_n \to x_*$.

Moreover, x_* is a fixed pt of T since

$$T(x_*) = T\left(\lim_{n\to\infty} x_n\right) \underset{\substack{\uparrow \\ \text{T is continuous}}}{=} \lim_{n\to\infty} T(x_n) = \lim_{n\to\infty} x_{n+1} = x_* . \quad \checkmark$$

Finally, to show that x_* is the <u>unique</u> fixed pt of T, let $y_* \in X$ be <u>any</u> fixed pt of T.

$$d(x_*, y_*) = d(T(x_*), T(y_*)) \leq L\, d(x_*, y_*),$$

So
$$d(x_*, y_*) \leq \underset{\substack{\uparrow \\ \in [0,1)}}{L}\, d(x_*, y_*)$$

$$\Longrightarrow \quad d(x_*, y_*) = 0 \Longrightarrow x_* = y_*. \quad \checkmark \qquad \square$$

<u>Def</u> Let (M,d) be a metric space, $x \in M$, $r \geq 0$. The open ball of radius r centered at x is

$$B_r(x) = B(r,x) = \{y \in M : d(x,y) < r\}.$$

The closed ball of radius r centered at x is

$$\overline{B}_r(x) = \overline{B}(r,x) = \{y \in M : d(x,y) \leq r\}.$$

Def Let (M,d), (N,ρ) be metric spaces. Let $A \subset M$. We say $T: M \to N$ is Lipschitz on A w/ constant $L \geq 0$ if

$$\forall a_1, a_2 \in A: \rho(T(a_1), T(a_2)) \leq L\, d(a_1, a_2).$$

We say that $T: M \to N$ is locally Lipschitz if $\forall x \in M: \exists \delta > 0:$ T is Lipschitz on $B_\delta(x)$.

Recall Let (X,d) be a metric space and $A \subset X$ be a subset. An open cover of A is a collection $\mathcal{U}$ of open sets $U \subset X$ s.t. $\bigcup_{U \in \mathcal{U}} U \supset A$.

The set A is compact if any open cover $\mathcal{U}$ of A has a finite subcover, i.e., $U_1, \dots, U_k \in \mathcal{U}$ s.t. $\bigcup_{i=1}^{k} U_i \supset A$.

The set A is sequentially compact if for any sequence (x_n) in A, $\exists$ a subsequence (x_{n_k}) that converges to some point <u>in A</u> as $k \to \infty$.

It is a fact that any set A is seq cpct $\Leftrightarrow$ it is compact.

Lem Let (M,d), (N,ρ) be metric spaces and $T: M \to N$ be locally Lipschitz. Then for any <u>compact</u> subset $A \subset M$, T is Lipschitz on A.

Pf Consider any $a \in A$. Since T is loc. Lip., $\exists \delta_a$ s.t. T is Lipschitz on $B(\delta_a, a)$ w/ const. $L_a \geq 0$.
Note that $\{B(\frac{\delta_a}{2}, a)\}_{a \in A}$ is an open cover of A. Since A is compact, $\exists a_1, \ldots, a_k \in A$

s.t. $A \subset \bigcup_{i=1}^{k} B\left(\frac{\delta_{a_i}}{2}, a_i\right)$.

Define $\tilde{L} := \max\{L_{a_1}, \ldots, L_{a_k}\}$ and $\delta := \min\{\delta_{a_1}, \ldots, \delta_{a_k}\}$.

Consider any $x, y \in A$.

<u>Case 1</u> $\exists\, i \in \{1, \ldots, k\}$ s.t. $x, y \in B\left(\frac{\delta_{a_i}}{2}, a_i\right)$.

Then $\rho(T(x), T(y)) \leq L_{a_i} d(x,y) \leq \tilde{L}\, d(x,y)$.

<u>Case 2</u> $\forall i \in \{1, \ldots, k\}$: $x, y \notin B\left(\frac{\delta_{a_i}}{2}, a_i\right)$.

Then $d(x,y) \geq \frac{\delta}{2}$ by the triangle inequality

(indeed, $\exists i$ s.t. $x \in B\left(\frac{\delta_{a_i}}{2}, a_i\right)$, so if $d(x,y) < \frac{\delta}{2}$ then

$d(y, a_i) \leq d(y,x) + d(x, a_i) < \frac{\delta}{2} + \frac{\delta_{a_i}}{2} \leq \delta_{a_i}$ and hence $y \in B\left(\frac{\delta_{a_i}}{2}, a_i\right)$,

contradicting that we are in Case 2).

Since A is cpct, the extreme value thm implies that

$$K := \sup_{a,b \in A} \rho(T(a), T(b)) = \max_{a,b \in A} \rho(T(a), T(b))$$ exists and is finite.

Thus, $$\rho(T(x), T(y)) \leq K = \frac{K}{d(x,y)} d(x,y) \leq \frac{2K}{\delta} d(x,y).$$

Thus, T is Lipschitz on A w/ constant

$$L = \max\left\{\tilde{L}, \frac{2K}{\delta}\right\}.$$

$\square$